

1 Solution — GIAM 2.1

Problem 2

1.1 Problem

Consider the sentence “This is a sentence which does not refer to itself.” (given at the start of the chapter). Is this sentence a statement? If so, what is its truth value?

1.2 Answer

Yes — it **is** a statement, and its truth value is **FALSE**.

The sentence *does* refer to itself (through the word “This”), so its claim that it does *not* refer to itself is simply false. Crucially — unlike the classic liar paradox — calling it false leads to **no contradiction**, so it qualifies as a legitimate (if self-defeating) statement.

1.3 The Reasoning

Call the sentence S . A **statement** is a declarative sentence that has a definite, consistent truth value. Let us see whether S can be assigned one.

Is the sentence consistently assignable a truth value? Follow the reasoning.

S: "This is a sentence which does not refer to itself."

Fact (about grammar, not truth): the word "This" points at S, so
S really does refer to itself.



What S asserts: "S does *not* refer to itself."
The assertion is the **opposite** of the fact above.



S asserts something false $\rightarrow$ S is FALSE.

Stability check — no paradox:

"S is false" means "S refers to itself is true" — which matches the fact.
Consistent. So S is a genuine statement with truth value FALSE.

Contrast — the Liar: "This sentence is false."
true $\Rightarrow$ false, and false $\Rightarrow$ true — an endless loop, no consistent value.
The Liar is NOT a statement. Our sentence is different: it settles on FALSE.

Figure 1.1: A reasoning chart. The sentence S is ‘This is a sentence which does not refer to itself.’ Step 1, a fact about grammar not truth: the word ‘This’ points at S , so S really does refer to itself. Step 2, what S asserts: ‘ S does not refer to itself’ — the opposite of the fact. Step 3, conclusion: S asserts something false, so S is FALSE. Step 4, stability check: ‘ S is false’ means ‘ S refers to itself is true’, which matches the fact — consistent, so S is a genuine statement with truth value FALSE. Contrast box, the Liar: ‘This sentence is false’ loops true-to-false-to-true endlessly, so it is not a statement; our sentence is different because it settles on FALSE.

Step 1 — a fact about reference. The demonstrative “This” in S points at S itself. Whether a sentence refers to itself is a matter of *grammar and reference*, settled independently of the sentence’s truth. And here the fact is clear: S **does refer to itself**.

Step 2 — what S claims. S asserts " S does *not* refer to itself" — the exact opposite of the Step 1 fact.

Step 3 — the verdict. Since S asserts the negation of something true, S asserts a falsehood. So S is false.

Step 4 — no paradox (the crucial check). Assigning “false” must not blow up. If S is false, then " S does not refer to itself" is false, i.e. " S *does* refer to itself" is true — which is exactly the Step 1 fact. Everything agrees. The value **false** is stable, so S is a bona-fide statement.

1.4 Remark — Why It Is *Not* a Paradox (Contrast With the Liar)

It is tempting to lump this in with the liar sentence “This sentence is false,” but they behave completely differently. The liar is genuinely paradoxical: assume it true and it must be false; assume it false and it must be true — no consistent value exists, so the liar is *not* a statement. Our sentence avoids the loop because its self-referential claim is about a **grammatical property** (does it refer to itself?), whose answer is fixed *before* we ever ask about truth. The liar’s claim, by contrast, is about **its own truth value**, which is precisely what we are trying to pin down — that circularity is what breaks it. So *S* is merely *self-defeating* (false), while the liar is *self-undermining* (valueless).

1.5 Remark — A Self-Refuting but Consistent Statement

S belongs to a tidy little family: sentences that are false *because* of what they say about themselves, yet perfectly consistent. Compare “This sentence has five words” (it has six, so it is false — no paradox) or “This sentence is written in French” (false, stated in English). Each refers to itself, makes a checkable claim about itself, and comes out false without any vicious circle. *S* is the same: it truthfully *could* have said “This sentence refers to itself” (which would be **true**), but by asserting the negation it lands on **false**. Self-reference alone does not create paradox — only self-reference *about truth* does.

1.6 Remark — Statements vs. Non-Statements in Logic

The exercise is really drilling the chapter’s central definition: a **statement** (or proposition) is a sentence that is definitely *either* true *or* false. Several kinds of sentence fail this test and are excluded from logic’s domain: questions (“What time is it?”), commands (“Shut the door.”), open sentences with free variables (“ $x > 3$ ” — neither true nor false until x is fixed), and genuine paradoxes (the liar). This problem sharpens the boundary by presenting a sentence that *looks* like it might be excluded — it is weird and self-referential — yet on inspection has a perfectly determinate truth value and so *is* admitted as a statement. The lesson: strangeness or self-reference does not disqualify a sentence; only the absence of a consistent truth value does.

1.7 Remark — Self-Reference Is Not Always Harmful

Self-reference has a fearsome reputation because of the liar and Russell’s paradox, but this problem shows it is not intrinsically toxic. A self-referential sentence can be true (“This sentence contains the word *sentence*”), false (our S), or paradoxical (the liar) — the outcome depends entirely on *what* the sentence says about itself. Productive self-reference is in fact essential to mathematics: Gödel’s incompleteness theorems are built by constructing a carefully controlled self-referential statement (“this statement is not provable”) that, unlike the liar, is consistent and turns out to be *true but unprovable*. Learning to tell harmless self-reference from paradoxical self-reference — exactly the discrimination this problem asks for — is the first step toward those deeper results.